

Gerard Bartolome

SENIOR STAFF / PRINCIPAL DATA PLATFORM ENGINEER | LAKEHOUSE | STREAMING | AWS

Los Angeles, CA | Remote / Hybrid | github.com/polyglotDataNerd

EXECUTIVE SUMMARY

Senior Staff / Principal-level Data Platform Engineer with 15+ years building cloud data platforms, real-time ingestion systems, lakehouses, and analytics infrastructure. Hands-on technical leader across **AWS and open-source data infrastructure**, including Apache Kafka, Spark Structured Streaming, Apache Iceberg, Delta Lake, Trino, dbt, Airflow, Kubernetes, Terraform, Go, Scala, Python, Rust, and SQL. Leads warehouse-to-lakehouse migrations, CDC frameworks, platform standards, data quality, observability, and production engineering for high-scale environments.

CORE PLATFORM EXPERTISE

AI-Augmented Engineering: Agentic development with Claude Code and Codex; automated code and PR review; custom Staff/Principal coding standards, review rubrics, and production-readiness checks; prompt/context engineering; LLM-assisted data-platform tooling.

Lakehouse & Analytics: Open table formats, Apache Iceberg, Delta Lake, Trino, PrestoDB, dbt, Spark, Redshift, Snowflake, ClickHouse, AWS Glue, S3, EMR

Streaming & Ingestion: Kafka / Amazon MSK, Spark Structured Streaming, CDC pipelines, real-time ingestion, Airflow / MWWA, AWS DMS, Kinesis, Lambda

Cloud & Infrastructure: AWS, Kubernetes / EKS, Spark on Kubernetes, ECS Fargate, Docker, Terraform, Helm, Prometheus, CloudWatch, DynamoDB, PostgreSQL, MongoDB Atlas, Redis

Languages & Architecture: Go, Scala, Python / PySpark, SQL, Java, Rust, TypeScript; open-source big data stack, lakehouse architecture, medallion architecture, event-driven systems, Kimball modeling, PII / CCPA, observability

SELECTED PLATFORM IMPACT

- Designed a Kafka-centric ingestion platform spanning push and pull APIs, Spark Structured Streaming, and Apache Iceberg on S3.
- Led warehouse and lakehouse migrations across Snowflake, Redshift, PrestoDB / Trino, Delta Lake, and Iceberg platforms.
- Built reusable Go, Scala, Spark, Terraform, and Kubernetes components for ingestion, CDC, orchestration, observability, and data modeling.
- Presented AWS re:Invent 2019 session **STG359-R: [Implementing a Data Lake on Amazon S3 ft. sweetgreen](#)**.

PROFESSIONAL EXPERIENCE

FloQast | Senior Staff Data Engineer Jan 2026 - Present

Platform: AWS S3, MSK / Kafka, EMR, ECS Fargate, Lambda, DynamoDB, Glue, MWWA, Spark 3.5 Structured Streaming, Trino, Iceberg, dbt, Airflow, MongoDB Atlas, Go, Scala, Python, Terraform

- **Designed agentic development and automated PR-review workflows with Claude Code and Codex**, encoding architecture standards and production-readiness rubrics across CDC, data modeling, observability, and EMR-to-ECS migration.
- **Own architecture for an event-driven, Kafka-centric API ingestion framework** that unifies push and pull paths through Spark Structured Streaming into an Iceberg lakehouse.
- Implemented Raw-to-Bronze-to-Silver-to-Gold medallion architecture on Iceberg / S3 with AWS Glue Catalog and dbt modeling on Trino and Spark.
- Designed MongoDB Atlas-to-S3 CDC through AWS DMS using Postgres-backed cursors, watermarks, and monotonic guards for idempotent, replay-safe ingestion.
- Built reusable DMS Terraform infrastructure with CloudWatch alarms, event subscriptions, incident automation, dashboards, and production observability.
- Developed Go ingestion gateways with bounded worker pools and Scala streaming consumers with deduplication, DLQ routing, partitioned Iceberg writes, and raw-archive forensics.

Ava Labs | Lead Staff Data Engineer Dec 2022 - Jan 2026

Platform: AWS S3, Kafka, Kubernetes, EMR, Spark 3.4, Trino, ClickHouse, Airflow, dbt, Delta Lake, Go, Scala, Python, Terraform, Helm

- **Architected migration from Snowflake to an internal S3 / Trino lakehouse**, reducing warehouse dependency for high-volume blockchain analytics.
- **Migrated Apache Spark workloads from EMR to Kubernetes**, including Spark-on-Kubernetes deployment patterns, Helm-based infrastructure, resource tuning, and production streaming operations.
- Built Kafka, Airflow, and Spark infrastructure for batch and streaming ingestion across Avalanche chains, subnets, and EVM-based networks.
- Developed a reusable Go WebSocket worker pool for multi-chain event ingestion into Kafka and an orchestration layer for real-time lakehouse analytics.
- Created shared Spark utilities for PII cleansing, Delta Lake I/O, and structured streaming; delivered an end-to-end Avalanche pipeline to Snowflake Marketplace.

PROFESSIONAL EXPERIENCE - CONTINUED

thredUP | Lead Staff Data Engineer Nov 2020 - Apr 2022

Platform: AWS S3, Kafka, EKS, EMR, Redshift, Aurora, Spark 3.1, Databricks, PrestoDB, Delta Lake, Airflow, Kubernetes, Terraform, Helm

- **Architected migration from a Redshift data warehouse to an S3 / Delta lakehouse** and served as SME for PrestoDB on Kubernetes.
- Designed Kimball-based analytical models spanning MPP warehouse and cloud lakehouse workloads.
- Built containerized ingestion microservices and deployed them to EKS through Helm and Jenkins.
- Developed orchestration to convert Redshift data to Apache ORC for PrestoDB and a shared Spark library for PII cleansing and streaming.

sweetgreen | Principal Big Data Engineer / Architect Feb 2018 - Mar 2020

Platform: AWS Kinesis, Lambda, API Gateway, EMR, Redshift, RDS, SQS, Spark 2.4, PostgreSQL, Redis, DynamoDB, ECS Fargate, Terraform, Go, Scala, Python

- **Designed and built a cloud data lake and event-driven analytics platform** supporting store operations, digital experiences, personalization, and CRM.
- Created a CCPA framework for PII anonymization and ad hoc deletion from the object store.
- Built open-source ingestion microservices and deployed containerized services to ECS Fargate with Terraform.
- Delivered serverless clickstream ingestion into Spark and Redshift plus an optimization application forecasting ingredient demand at 15-minute intervals.

Beachbody On Demand | Big Data Engineer / Architect Oct 2015 - Feb 2017

Platform: AWS Kinesis, Lambda, API Gateway, EMR, Redshift, RDS, SQS, Hadoop, Spark, Cassandra, DynamoDB, Java, Scala, Python

- Built a **terabyte-scale AWS data platform** integrating multiple sources for event-driven personalization, analytics, and user experience.
- Developed serverless real-time clickstream ingestion into Hadoop and Redshift for recommendations and analytics.
- Built a Spark ML collaborative-filtering personalization engine from user workout behavior and an Alexa telemetry pipeline on AWS serverless services.

NBCUniversal / Fandango | Data Engineer / Architect Aug 2013 - Sep 2015

Platform: AWS, Hadoop, Spark, Redshift, Tableau, PostgreSQL, MySQL, SQL Server, Java, Scala, Python

- Integrated on-premises data centers with AWS to build a hybrid data lake and terabyte-scale analytics platform.
- Implemented and monitored end-to-end ETL / ELT pipelines supporting insights, machine learning, forecasting, and predictive modeling.

EARLIER EXPERIENCE

Stylesight Inc. | Data Warehouse / Business Intelligence Architect | Feb 2012 - 2014

Designed Kimball star schemas and SQL Server ETL for Salesforce, Omniture, and Marketo; owned database performance, administration, and BusinessObjects analytics delivery.

MTV Networks | Data Warehouse Architect / Developer | Jan 2010 - 2012

Led SQL Server and Oracle warehouse development for forecast, budget, royalty, cash, and invoice reporting; built ETL, database logic, optimization, and BusinessObjects semantic models.

EDUCATION & RECOGNITION

Louisiana State University, Baton Rouge, LA | Bachelor of Science, Computer Science / Managerial Information Science

AWS re:Invent 2019 Speaker - STG359-R | [Implementing a Data Lake on Amazon S3 ft. sweetgreen](#)